TROPIC01

ODN_TR01_int_guide

TROPIC01 integration guide

Version: 1.0

Abstract

This document outlines recommended practices for the streamlined integration of TROPIC01 into customer products.

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Contents

1 Introduction **2**

2 Integration Guidelines **3**
2.1 TROPIC01 Features description 3
2.2 Hardware Integration 3
2.3 Software TROPIC01 integration Workflow overview 4
2.4 Software Support and SDK 4
2.5 Ordering, Packaging, and Labeling 5
2.6 Manufacturing and Provisioning 5

3 Lifecycle and Support **6**
3.1 Product Lifecycle and Change Management 6
3.2 Security Lifecycle 6
3.3 Customer support 6

1 Introduction

This document serves as a starting point for developers integrating TROPIC01 into a new product. It lists all important steps in integration and provides navigation to the technical documentation, supporting software, and hardware resources required at every stage of the development.
The target audience for this document includes:

- **System Architects** - For a high-level overview of available integration resources.
- **Hardware Engineers** - For specific physical and electrical integration requirements.
- **Embedded Software Engineers** - For details on supporting software, drivers, and feature implementation.
- **Operations and Manufacturing** - For guidance on TROPIC01 handling, packaging, and marking.

2 Integration Guidelines

This chapter lists areas which are important during the integration process. By following these guidelines, developers can ensure that TROPIC01 is implemented according to Tropic Square recommendations, reducing design iterations and accelerating time-to-market.

2.1 TROPIC01 Features description

The Datasheet [1] provides a high-level functional overview, while specific security and configuration features are detailed in Application Notes. For communication protocol specifics, refer to the User API documentation [2].

- **Device Identity** [4] - The TROPIC01 PKI infrastructure.
- **Pin Verification (MACANDD)** [3] - Patent pending end user authentication.
- **Configuration Objects** [6] - Device configuration and behavioral effects.
- **First Pairing key** [5] - Default SH0 pairing key and ownership change mechanism.
- **Firmware update** [7] - TROPIC01 Firmware update mechanism.

2.2 Hardware Integration

The TROPIC01 interfaces with Host MCU as an **SPI slave device**. The electrical and physical constraints are defined in the TROPIC01 Datasheet [1]. Key integration parameters and requirements are summarized in the following sections of the Datasheet:

- **Reference Schematic** - See *TYPICAL APPLICATION* [1] for standard SPI wiring and decoupling requirements.
- **Package and PCB Layout** - See *PCB LAND PATTERN* and *PACKAGE DESCRIPTION* [1].
- **Soldering Guidelines** - See *SOLDERING RECOMMENDATIONS* [1].
- **Electrical Specifications** - See *ELECTRICAL SPECIFICATIONS* and *ABSOLUTE MAXIMUM RATINGS* [1] for SPI timing characteristics and voltage levels.

2.3 Software TROPIC01 integration Workflow overview

Due to the permanent nature of TROPIC01 configurations, we recommend a staged integration workflow to streamline the early phases of development:

1. **Simulation** - Begin with the TROPIC01 Python model [19]. This provides a risk-free environment to explore device behavior specifically regarding ownership transfer and *I-Config*/*R-Config* impacts without the danger of permanently locking physical hardware. Refer to Libtropic Model Tutorials [15] for specific instructions how to use the model with Libtropic.
2. **Prototyping** - Transition to TROPIC01 Development Boards once the functional logic is verified in simulation. This stage facilitates real-world electrical, timing and functional validation.
3. **Final Integration** - Proceed to full integration into the customer product only after the prototyping phase has been successfully validated on Dev Boards.

2.4 Software Support and SDK

To accelerate integration to customer product, Tropic Square provides **Libtropic** [13], a production-ready C-language SDK designed for seamless integration into resource-constrained environments.

- **Platform Support** - Ready-to-use abstractions for Linux, STM32, ESP32, Arduino
- **Implementation** - Optimized "bare metal" C for minimal memory footprint and maximum portability.
- **Code Stability** - Production ready, support and maintenance provided directly by Tropic Square.

Note

All available tutorials for **Libtropic**:
`https://tropicsquare.github.io/libtropic/latest/tutorials/`

2.5 Ordering, Packaging, and Labeling

All logistics and procurement details are managed via the TROPIC01 Product Catalog [18]. This document serves as the authoritative reference for the following:

- **Part Number (P/N) Nomenclature** - A detailed breakdown of the ordering codes, including silicon revision and package variants.
- **Device Marking** - Explanation of the markings on the physical TROPIC01 package.
- **Packaging Specifications** - Details regarding Tape and Reel dimensions, moisture barrier bags, and standard box quantities.
- **Moisture Sensitivity Level (MSL)** - The MSL rating and associated dry-pack requirements for surface-mount assembly.
- **Ordering Information** - A list of currently active and orderable Part Numbers.

Note

For direct procurement of Dev Boards or standalone TROPIC01 Devices please visit:
`https://www.tropicsquare.com/order-devkit`

2.6 Manufacturing and Provisioning

During the mass production of a product containing TROPIC01, the following mandatory steps shall be implemented:

- **Device Identity Verification** - Consult the *PKI Application Note* [4] to implement authenticity checks, ensuring only genuine TROPIC01 devices are used in production.
- **Ownership Transfer** - Follow the *Chip Ownership* procedures outlined in the Datasheet [1] to secure the device. For a practical implementation and code examples, refer to the *Default Pairing Keys for a Secure Channel Handshake* [16] section in Libtropic documentation.
- **Firmware Baseline** - Ensure all devices are flashed with the latest firmware version using the TROPIC01 update mechanism described in *FW Update Application Note* [7]. For a practical implementation and code examples, refer to the *TROPIC01 Firmware* [17] section in the Libtropic documentation.

Recommended Optional Steps:

- **Configuration Audit** - Evaluate if factory-default *Configuration Objects* require modification to align with specific product security or performance requirements.

3 Lifecycle and Support

3.1 Product Lifecycle and Change Management

Tropic Square maintains a transparent disclosure policy regarding hardware errata and manufacturing modifications. We categorize updates into two main streams to help your team prioritize impact:

- **Errata** - Documentation of identified product flaws that may adversely impact functionality.
- **Product Change Notices (PCN)** - Notifications regarding changes in manufacturing, device marking, or breaking changes in firmware.

Relevant updates are communicated through the following official channels:

- **Errata Log** [8] - Cumulative list of all known device errors.
- **PCN Repository** [9] - Formal notices regarding supply chain, marking, or functional behavioral shifts.

3.2 Security Lifecycle

The TROPIC01 security features, attacker model and adressed threats are described in the **Security Architecture Application Note** [12].
Tropic Square works with its partners to validate security properties of TROPIC01. All TROPIC01 public validation **Security Reports** are available online [11].
Tropic Square maintains a transparent disclosure policy regarding severe TROPIC01 weaknesses and vulnerabilities. The known vulnerabilities of TROPIC01 are covered by **Security Advisories** (SA). All TROPIC01 SAs are available online [10].

3.3 Customer support

We are committed to support TROPIC01 if any question arises, please contact us via our support desk (requires login).

Version history

Version	Date	Description
1.0	05.11.2026	The first public version of the integration guide.

References

[1] **TROPIC01 Datasheet**, Tropic Square
Document Name: ODD_TR01_datasheet
GitHub: **See note below**

[2] **TROPIC01 User API**, Tropic Square
Document Name: ODU_TR01_user_api
GitHub: **See note below**

Note

The TROPIC01 is behavior differs between Part Numbers (P/N), always refer to datasheet and corresponding User API based on P/N! The combination list is available on GitHub: `https://github.com/tropicsquare/tropic01/blob/main/doc/pages/part-numbers.md`

[3] **Pin Verification Application Note**, Tropic Square
Document Name: ODN_TR01_app_002
GitHub: `https://github.com/tropicsquare/tropic01/blob/main/doc/application_notes/ODN_TR01_app_002_pin_verif_1v2.pdf`

[4] **Device Identity and PKI Application Note**, Tropic Square
Document Name: ODN_TR01_003
GitHub: `https://github.com/tropicsquare/tropic01/blob/main/doc/application_notes/ODN_TR01_app_003_pki_1v3.pdf`

[5] **First Pairing Key Application Note**, Tropic Square
Document Name: ODN_TR01_app_005
GitHub: `https://github.com/tropicsquare/tropic01/blob/main/doc/application_notes/ODN_TR01_app_005_first_pairing_key_1v3.pdf`

[6] **Configuration Objects Application Note**, Tropic Square
Document Name: ODN_TR01_app_006
GitHub: `https://github.com/tropicsquare/tropic01/blob/main/doc/application_notes/ODN_TR01_app_006_config_obj_1v3.pdf`

[7] **FW Update Application Note**, Tropic Square
Document Name: ODN_TR01_app_007
GitHub: https://github.com/tropicsquare/tropic01/blob/main/doc/application_notes/ODN_TR01_app_007_fw_update_1v5.pdf

[8] **TROPIC01 Erratas**, Tropic Square
GitHub: https://tropic-ops.atlassian.net/wiki/external/MjRmZDY4ZWNlMTYzNGIwZWJkZGQwNzlhM2U5NWQzN2E

[9] **TROPIC01 Product Change Notifications**, Tropic Square
GitHub: https://tropic-ops.atlassian.net/wiki/external/ODVlMjc1ZWQzM2YxNDEwNmJlZGY0NTk4YWIxMDg0NDI

[10] **TROPIC01 Security Advisories**, Tropic Square
GitHub: https://github.com/tropicsquare/tropic01/blob/main/doc/pages/security-advisories.md

[11] **TROPIC01 Security Reports**, Tropic Square
GitHub: https://github.com/tropicsquare/tropic01/blob/main/doc/pages/security-reports.md

[12] **Security Architecture Application Note**, Tropic Square
Document Name: ODN_TR01_app_008
GitHub: https://github.com/tropicsquare/tropic01/blob/main/doc/application_notes/ODN_TR01_app_008_sec_arch_1v1.pdf

[13] **TROPIC01 SDK** – Libtropic, Tropic Square
GitHub: https://github.com/tropicsquare/libtropic

[14] **TROPIC01 SDK - Documentation** – Libtropic, Tropic Square
GitHub: https://tropicsquare.github.io/libtropic/latest/

[15] **TROPIC01 SDK Documentation - Model Tutorials** – Libtropic, Tropic Square
GitHub: https://tropicsquare.github.io/libtropic/latest/tutorials/model/

[16] **TROPIC01 SDK Documentation - Default Pairing Keys for a Secure Channel Handshake** – Libtropic, Tropic Square
GitHub: https://tropicsquare.github.io/libtropic/latest/reference/default_pairing_keys/

[17] **TROPIC01 SDK Documentation - TROPIC01 Firmware** – Libtropic, Tropic Square
GitHub: https://tropicsquare.github.io/libtropic/latest/reference/tropic01_fw/

[18] **TROPIC01 Catalog**, Tropic Square
GitHub: `https://github.com/tropicsquare/tropic01/blob/main/doc/catalog_list/OD_TR01_catalog_list_2v0.pdf`

[19] **TROPIC01 Model**, Tropic Square
GitHub: `https://github.com/tropicsquare/ts-tvl`

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